Abstract

These notes comprise a scribed record of the lectures for the *Algorithms* Course conducted at the TIFR during the August–November 2024 semester, instructed by Prof. Umang Bhaskar. The official course webpage is available at this link.

The primary reference for the course was *Introduction to Algorithms* by Cormen, Leiserson, Rivest, and Stein (CLRS), complemented by supplementary materials and the lecturer's own notes, which are accessible through the course webpage. For the chapter on approximation algorithms, chapters 14, 15, and 17 of *Approximation Algorithms* by Vijay Vazirani were also used.

These notes were prepared both as a personal academic resource and to facilitate revision for other students. They have been typeset using a custom theme developed by me. The source code of the notes is available at this repository, and the source code of the theme can be found in my GitHub repository. Any corrections, comments, or suggestions are welcome and reach out to me at `soham.chatterjee@tifr.res.in`.

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Chapter 1

Introduction

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Some Basics of Finite Fields

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Decoding from Quadratic Character Evaluations